

UEC645: MICROWAVE ENGINEERING				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: To enhance student's comprehensive capabilities in Microwave engineering through understanding of electromagnetic wave generation, transmission and measurements theory and technology by study of microwave transmission medium, media and microwave devices and components.				
Syllabus Wave Guides and Resonators: TE, TM Modes in rectangular & Circular wave guides, characteristics impedance of waveguides, Excitation of wave guides. Microwave Components: Waveguide Microwave Junctions, Scattering matrix and their properties, Microwave T junctions – H Plane Tee, E Plane Tee, Directional coupler – Two hole directional coupler and scattering matrix of a directional coupler, Waveguide joints, Bends, Corners, Transition & twists, Waveguide terminations, Ferrite devices – faraday rotation in devices, Circulator & isolator. Microwave Tubes and Circuits: High frequency limitations of conventional tubes, Klystrons - two cavity klystron amplifier & oscillator, Reflex klystron, Travelling wave tube, Microwave cross-field tube magnetron – Operation and MW characteristics. Microwave Measurements: General measurement setup, Microwave bench, Power measurement – low, Medium & high, Attenuation measurement, Measurement of VSWR, Measurement of dielectric constant, Measurement of Impedance: using Smith Chart, Measurement with spectrum analyzer, Scalar & vector network analyzer operation. Microwave Solid State Devices & Their Applications: P-I-N devices, GUNN Diode, IMPATT. Familiarization with standards: IEEE 521-2002, IEEE 147-1979, IEEE 470-1972 and IEEE 544-1975, IRE 2.S1-1955				
Laboratory Work: To study the performance of mode characteristics of reflex klystrons circulator, Characteristics of Gunn diode, Directional coupler, Attenuator, Sliding screw tuner, Verify the relation of wavelength, finding unknown impedance, VSWR measurement, E-plane, H-plane, Magic Tee, Computer based simulation experiments.				
Course Learning Objectives (CLO) The students will be able to: 1. Analyse different propagating modes in a waveguide 2. Examine various microwave components and their operation. 3. Explain the basic principles of devices used in microwave generation. 4. Evaluate the microwave measurement techniques for various microwave components.				
Text Books 1. Liao, S.Y., <i>Microwave Devices & Circuits</i>, Tata McGraw Hill (2006) 2nd edition. 2. David M. Pozar, <i>Microwave Engineering</i>, Wiley-India(2011) 3rd edition 3. Collins, Robert, <i>Foundation of Microwave Engineering</i>, McGraw Hill (2005) 3rd edition.				

Reference Books

1. Wolf E.A., and kaul, R., *Microwave Engineering & Systems Applications*, Wiley Interscience (2002) 4th edition.
2. Sze, S. M., *Physics of Semiconductor Devices*, Wiley Eastern (2003) 2nd edition.
3. Sarvate, V.V., *Electromagnetic Fields & Waves*, John Wiley & Sons (2004) 3rd edition.